

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

Both the [Eloundou et al. \(2024\)](#) measures and the [Handa et al. \(2025\)](#) measures estimate AI exposure by 2018 SOC code. We use a 2010 SOC code to 2018 SOC code crosswalk from the BLS to merge the exposure measures to the payroll data. Table [A1](#) shows example occupations for each AI exposure measure.

3.3 Other Data

To compare employment changes for teleworkable versus non-teleworkable occupations, we use data from [Dingel and Neiman \(2020\)](#). We use the Personal Consumption Expenditure index from the BLS to compute real earnings, indexed to October 2017. We use monthly Current Population Survey (CPS) data as a comparison for our main findings.

4 Results

4.1 Fact 1: Employment for young workers has declined in AI-exposed occupations

Consider software engineers and customer service agents, two occupations that are frequently considered to be highly exposed to generative AI tools. Media attention has raised the specter of widespread employment disruption for young software engineers in particular ([Thompson, 2025](#); [Raman, 2025](#); [Allen, 2025](#); [Horowitch, 2025](#)).

Figure [1](#) shows employment changes by age group for these occupations, normalized to 1 in October 2022. Both occupations present a similar pattern: employment for the youngest workers declines considerably after 2022, while employment for other age groups continues to grow. By July, 2025, employment for software developers aged 22-25 declined by nearly 20% compared to its peak in late 2022. Figure [A1](#) shows that a similar pattern holds for computer occupations and service clerks more generally.

Figure [2](#) shows four other professions as case studies, spanning varying levels of AI exposure according to the measures developed in [Eloundou et al. \(2024\)](#). Marketing and sales managers, in

refinement process; Learning, meaning knowledge acquisition and understanding; Validation, meaning work verification and improvement; or “None,” with the model instructed to choose the None option “liberally.” Conversations classified as Directive or Feedback Loop are considered as Automative, while ones classified as Task Iteration, Learning, or Validation are considered Augmentative. See [Handa et al. \(2025\)](#) for more details. Table [A2](#) reproduces Table 1 from [Handa et al. \(2025\)](#) and shows more details about the automation and augmentation measures.

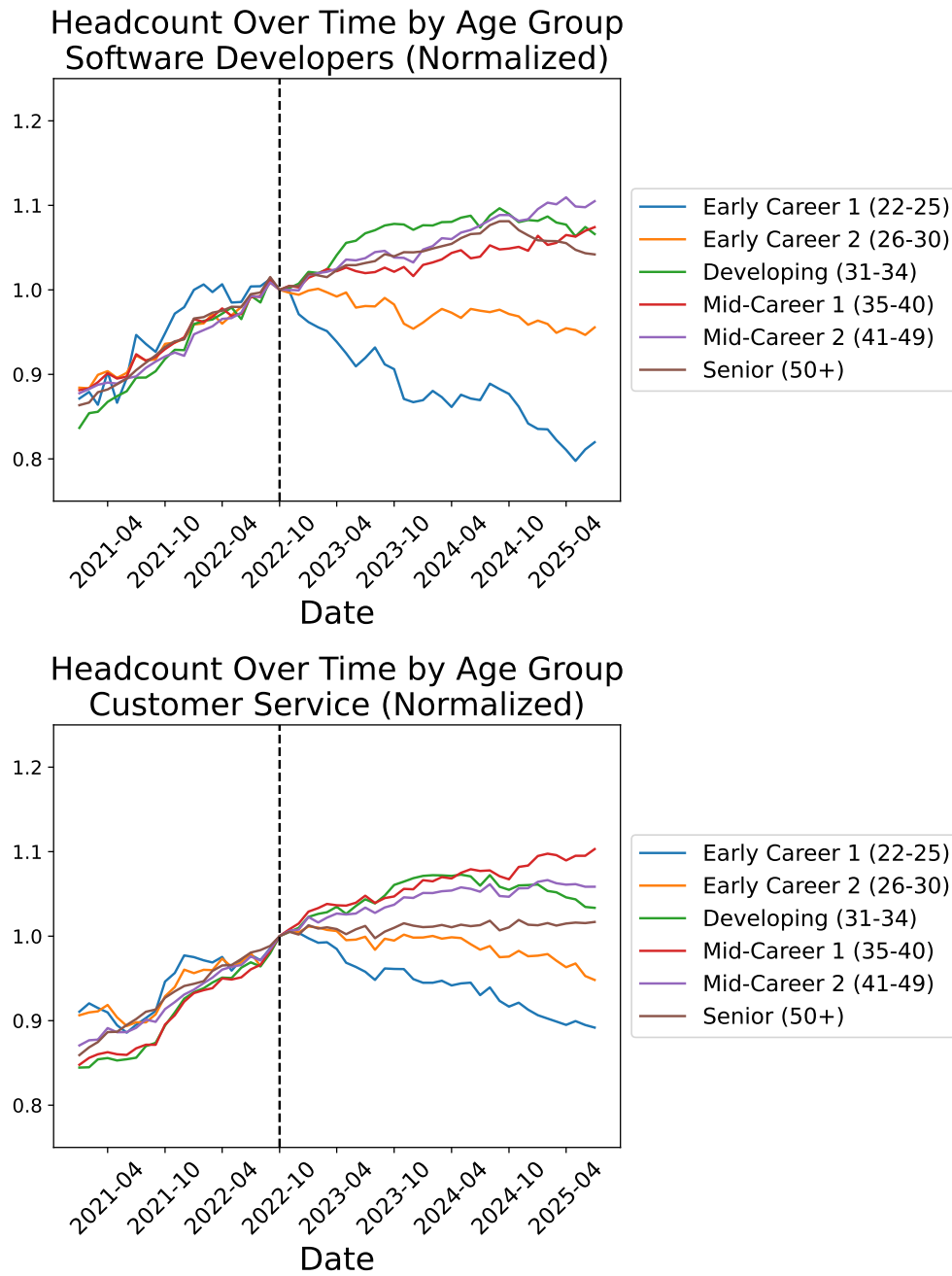


Figure 1: Employment changes for software developers and customer service agents by age, normalized to 1 in October 2022.

f indexes firms, q indexes Eloundou et al. (2024) exposure quintiles, and t indexes months, with $t = -1$ corresponding to October 2022. The outcome variable $y_{f,q,t}$ is employment in f, q, t . Equation 4.1 is a Poisson event study regression controlling for firm-quintile effects, $\alpha_{f,q}$, and firm-time effects, $\beta_{f,t}$. The firm-time effects absorb aggregate firm shocks that impact each exposure quintile equally. The firm-quintile effects adjust for baseline differences in hiring across quintiles within the firm. The coefficients of interest, $\gamma_{q,t}$, measure differential changes in employment growth across quintiles after accounting for firm-time effects and firm-quintile effects.¹⁷

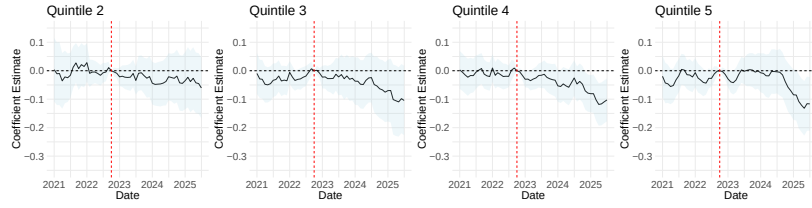
We run this regression separately for each age group. For each regression, we restrict to firms that hire at least 10 workers within the age group in every period of the sample. Further, $\sum_t y_{f,q,t}$ must equal at least 100 for each q , meaning that the firm must at least employ on average about 2 workers from each exposure quintile across months in the sample.¹⁸ Standard errors are clustered by firm.

Results are in Figure 9, which plots the $\gamma_{q,t}$ coefficients for each age group. For workers aged 22-25, estimates for higher quintiles are large and statistically significant, with a 12 log point decline in relative employment comparable in magnitude to the estimates in the raw data in Figure 3. Estimates for other age groups are generally much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

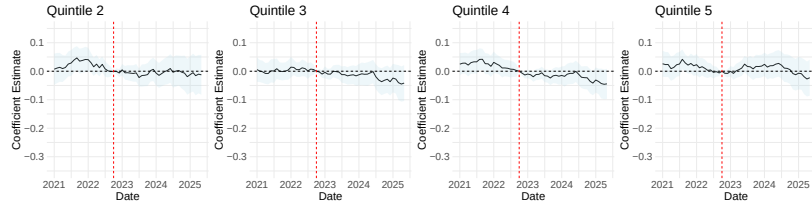
One alternative confounder that would not be controlled for with firm-time effects is that even conditional on the firm workers with high AI exposure were excessively hired after the COVID-19 pandemic, leading to a subsequent contraction in their hiring. To assess such alternatives we consider various other robustness checks in Section 4.6, such as removing computer occupations and conditioning on whether the occupation is amenable to work from home.

¹⁷Because of zero counts in the outcome variable, we estimate a Poisson regression instead of an OLS regression in logs following guidance from Chen and Roth (2024).

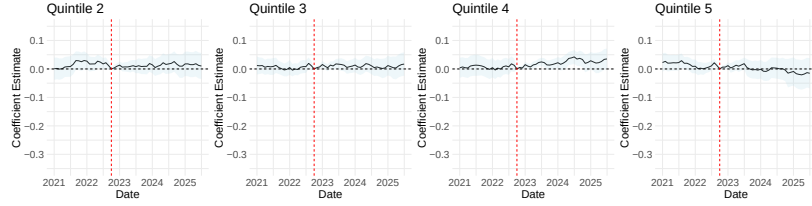
¹⁸Results are not sensitive to these restrictions, though there must be at least one non-zero value in each firm-month and each firm-quintile for observations to not get dropped in the Poisson regression.



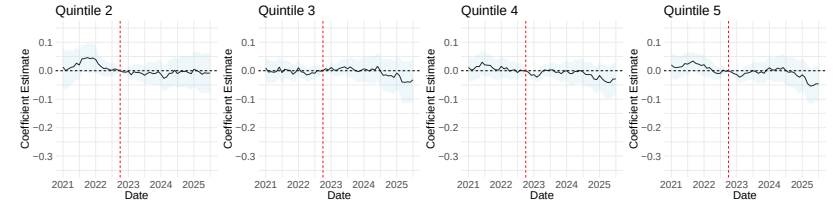
(a) Age 22-25



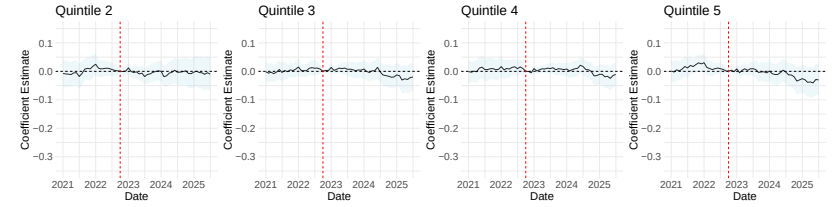
(c) Age 31-34



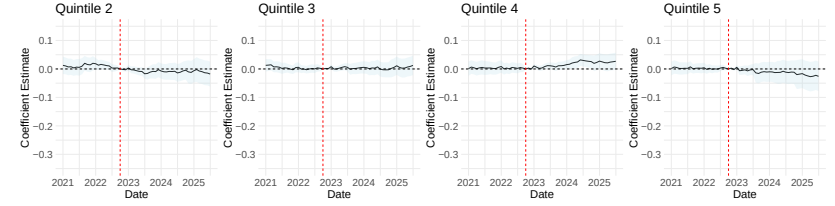
(e) Age 41-49



(b) Age 26-30



(d) Age 35-40



(f) Age 50+

Figure 9: Poisson regression event study estimates for employment changes by age and AI exposure. Estimates are all relative to occupational exposure quintile 1. Exposure quintiles use [Eloundou et al. \(2024\)](#) GPT-4 β measures. Estimates control for firm-time and firm-quintile fixed effects following Equation 4.1. Shaded regions are 95% confidence intervals. Standard errors are clustered by firm.